

Task 1: To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs

```
GNU nano 7.2                                                                    pr
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        printf("Child process\n");
        printf("Child PID: %d\n", getpid());
    } else {
        printf("Parent process\n");
        printf("Parent PID: %d\n", getpid());
        wait(NULL);
        printf("Child process completed\n");
    }

    return 0;
}
```

```
shreyas@Grigori:~/skill_session_5$ mkdir -p ~/skill_session_6
shreyas@Grigori:~/skill_session_5$ cd ~/skill_session_6
shreyas@Grigori:~/skill_session_6$ nano escape_test.sh
shreyas@Grigori:~/skill_session_6$ chmod +x escape_test.sh
shreyas@Grigori:~/skill_session_6$ ./escape_test.sh
Hello\ World
Hello "World"
Dollar: $HOME
Special: \# \! \*
Path: /home/Shreyas/My\ Folder
shreyas@Grigori:~/skill_session_6$ |
```

Task 2: To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching

```
shreyas@Grigori:~/skill_session_5$ cd ~/skill_session_6
shreyas@Grigori:~/skill_session_6$ nano escape_test.sh
shreyas@Grigori:~/skill_session_6$ chmod +x escape_test.sh
shreyas@Grigori:~/skill_session_6$ ./escape_test.sh
Hello\ World
Hello "World"
Dollar: $HOME
Special: \# \! \*
Path: /home/Shreyas/My\ Folder
shreyas@Grigori:~/skill_session_6$ nano process.c
shreyas@Grigori:~/skill_session_6$ gcc process.c -o process
shreyas@Grigori:~/skill_session_6$ ./process
Parent process
Parent PID: 427
Child process
Child PID: 428
Child process completed
shreyas@Grigori:~/skill_session_6$ nano exec_test.c
shreyas@Grigori:~/skill_session_6$ gcc exec_test.c -o exec_test
shreyas@Grigori:~/skill_session_6$ ./exec_test
Before exec
total 44
-rwxr-xr-x 1 shreyas shreyas 141 Sep 27 12:48 escape_test.sh
-rwxr-xr-x 1 shreyas shreyas 16048 Sep 27 12:50 exec_test
-rw-r--r-- 1 shreyas shreyas 164 Sep 27 12:50 exec_test.c
-rwxr-xr-x 1 shreyas shreyas 16176 Sep 27 12:50 process
-rw-r--r-- 1 shreyas shreyas 451 Sep 27 12:50 process.c
shreyas@Grigori:~/skill_session_6$ ls
escape_test.sh exec_test exec_test.c process process.c
shreyas@Grigori:~/skill_session_6$ |
```

```
#include <stdio.h>
#include <unistd.h>

int main() {
    printf("Before exec\n");

    execlp("ls", "ls", "-l", NULL);

    perror("exec failed");

    return 1;
}
```